Z-Menu Edit SOP

Standard Operating Procedure — Modifying zmenu.sh on the ZBook Ultra

Machine: HP ZBook Ultra 14 G1a · Ubuntu 24.04
Source file: `~/projects/zmenu/zmenu.sh`
Live binary: `/usr/local/bin/zmenu`
Current version: 5.1.0 (update this line each time you bump)

THE GOLDEN RULE

> Always edit the **source file** (`~/projects/zmenu/zmenu.sh`).
> Never edit `/usr/local/bin/zmenu` directly — it gets overwritten on install.

PHASE 1 — BEFORE YOU TOUCH ANYTHING

Step 1.1 — Backup

Run this every single time, no exceptions:

```bash
ZMENU_SRC=~/projects/zmenu/zmenu.sh
BACKUP_DIR="$(dirname "$ZMENU_SRC")/.zmenu-backups"
mkdir -p "$BACKUP_DIR"
cp "$ZMENU_SRC" "$BACKUP_DIR/zmenu-$(date +%Y%m%d-%H%M%S).sh"
cp /usr/local/bin/zmenu "$BACKUP_DIR/zmenu-installed-$(date +%Y%m%d-%H%M%S).sh"
ls -t "$BACKUP_DIR"/zmenu-2*.sh | tail -n +21 | xargs rm -f 2>/dev/null || true
echo "Backup done → $BACKUP_DIR"
```

Verify it worked:

```bash
ls -lh ~/.zmenu-backups/ 2>/dev/null || ls -lh ~/projects/zmenu/.zmenu-backups/
```

Step 1.2 — Check what version is live vs source

```bash
grep 'ZMENU_VERSION' ~/projects/zmenu/zmenu.sh | head -1
grep 'ZMENU_VERSION' /usr/local/bin/zmenu | head -1
```

Both should match. If they don't, the source is ahead of the installed copy — that's fine, just be aware.

Step 1.3 — Find the section you want to change

Method A — search by function name (fastest):

```bash
grep -n 'function_name\|_ai_ollama\|mod_docker' ~/projects/zmenu/zmenu.sh
```

Method B — search by visible menu text:

```bash
grep -n 'text you see on screen' ~/projects/zmenu/zmenu.sh
```

Method C — section map (know the architecture):

Section	Line range (approx)	What lives here
Config	~1–100	`ZMENU_VERSION`, defaults, colours
Discovery	~100–400	`discover()`, all `_disc_*`, `D_*` vars
Context generator	~400–550	`context_generate()`
AI launcher	~550–650	`cc_launch()`, `owui_check()`
Chrome / UI	~650–750	`header()`, `pause()`, `status_bar()`
Modules	~750–3100	All `mod_*` and `_module_action` fns
Main menu	~3100–3200	`main_menu()`
Entrypoint	~3200–end	`_bootstrap()`, CLI arg handler

Find exact line for any function:

```bash
grep -n '^_ai_ollama_settings\|^mod_docker\|^_sec_ports' ~/projects/zmenu/zmenu.sh
```

Jump to a line in nano:

```
Ctrl+_ then type the line number, Enter
```

Jump to a line in vim/nvim:

```
:1342   (then Enter)
```

PHASE 2 — MAKING THE EDIT

Step 2.1 — Open the source file

```bash
# nano (simplest):
nano ~/projects/zmenu/zmenu.sh

# or your preferred editor:
code ~/projects/zmenu/zmenu.sh
nvim ~/projects/zmenu/zmenu.sh
```

Step 2.2 — Find your target (in nano)

- `Ctrl+W` → type the function name or visible text → Enter
- `Ctrl+W` again to find next occurrence
- `Ctrl+_` → enter a line number to jump directly

Step 2.3 — Understand what you're editing

Every module follows the **same pattern**. Recognising it saves you from breaking things:

```bash
```

```bash
_ai_example_function() {              # ← function declaration
    while true; do                    # ← menu loop (present in menus, not in helper
        header                        # ← clears screen + prints banner
        echo -e "${BCYN}⋯ TITLE${NC}" # ← section header (cyan, always this style)
        echo ""
        # ... display state ...
        echo "   1)  Option one"      # ← menu options (3 spaces indent)
        echo "   2)  Option two"
        echo "   r)  Back"
        echo ""
        read -rp "  Selection: " ch   # ← input prompt
        case $ch in
            1) _ai_do_thing; pause ;; # ← calls helper, then pauses
            2) _ai_other; pause ;;
            r|R) break ;;             # ← always have this
            *) echo -e "${RED}  Invalid.${NC}"; sleep 1 ;;
        esac
    done
}
```

Helpers (no while loop, just do work and return):

```bash
_ai_do_thing() {
    header
    echo -e "${BCYN}⋯ DO THING${NC}"
    echo ""
    # ... do the work ...
    echo -e "  ${OK}  Done"
}
```

Step 2.4 — Colour / symbol quick reference

```bash
${BCYN}text${NC}    # Cyan  — section headers, highlighted values
${BOLD}text${NC}    # Bold  — field labels
${DIM}text${NC}     # Dim   — hints, secondary info, notes
${BRED}text${NC}    # Red   — destructive actions (Stop, Delete)
$OK               # Green ●  healthy / success
$FAIL             # Red ●    error / stopped
$WARN             # Yellow ● warning
$IDLE             # Grey ●   off / idle
```

Step 2.5 — Version bump rule

Find and update line ~33:

```bash
readonly ZMENU_VERSION="X.Y.Z"
```

Type of change	Example	Bump
Typo, cosmetic, wording	Fix a label	Patch: 5.1.0 → 5.1.1
New option in existing menu, new helper fn	Add a new setter	Minor: 5.1.0 → 5.2.0
New top-level module, new config var, arch change	Add a whole new section	Major: 5.1.0 → 6.0.0

PHASE 3 — SAVE AND INSTALL

Step 3.1 — Save the source file

- **nano:** `Ctrl+O` → Enter → `Ctrl+X`
- **vim:** `:wq`
- **VS Code:** `Ctrl+S`

Step 3.2 — Syntax check (always do this before installing)

```bash
bash -n ~/projects/zmenu/zmenu.sh && echo "✓ Syntax OK" || echo "✗ SYNTAX ERROR — do
```

If you get a syntax error it will tell you the line number. Fix it before proceeding.

Step 3.3 — Install to live path

```bash
sudo cp ~/projects/zmenu/zmenu.sh /usr/local/bin/zmenu && \
sudo chmod +x /usr/local/bin/zmenu && \
echo "✓ Installed"
```

Step 3.4 — Verify the version changed

```bash
```

```
grep 'ZMENU_VERSION' /usr/local/bin/zmenu | head -1
```

Step 3.5 — Run it

```bash
zmenu
```

PHASE 4 — WHEN THINGS GO WRONG

Problem: zmenu won't start / crashes immediately

```bash
# Run with bash's trace mode to see exactly what fails:
bash -x /usr/local/bin/zmenu 2>&1 | head -50
```

Look for the last line before the error — that's where it died.

Problem: You see "unbound variable" error

This means a variable is used before being set. `set -u` is active in zmenu.

Fix pattern:

```bash
# WRONG — crashes if VAR is not set:
echo "$SOME_VAR"

# RIGHT — safe with fallback:
echo "${SOME_VAR:-}"          # empty fallback
echo "${SOME_VAR:-default}"   # explicit fallback
```

Search for the variable name to find where it should be initialised.

Problem: A function silently does nothing / exits early

`set -e` is active — any command that returns a non-zero exit code kills the function.

Fix pattern:

```bash
```

```
# WRONG — if `somecommand` fails, whole script dies:
somecommand

# RIGHT — suppress failure:
somecommand 2>/dev/null || true
somecommand 2>/dev/null || echo "failed, continuing"
result=$(somecommand 2>/dev/null || echo "")
```

Problem: Menu option does nothing when pressed

Check three things in order:

1. **The `case` letter matches exactly** — `r` and `R` are different characters
2. **The function name is spelled correctly** — typos are silent in bash
3. **The function actually exists** — search for it:

```bash
grep -n '^_function_name' ~/projects/zmenu/zmenu.sh
```

Problem: Discovery data is stale / wrong

zmenu runs discovery once at startup. Force a re-run without restarting:

```bash
# From inside zmenu:
# Settings → Re-run discovery

# Or from terminal (restart zmenu):
zmenu
```

To debug a specific discovery function:

```bash
bash --norc -c 'source ~/projects/zmenu/zmenu.sh; discover; echo "GPU: $D_GPU_DRIVER
```

Problem: Something broke, need to roll back

```bash
# List your backups (newest first):
ls -lt ~/projects/zmenu/.zmenu-backups/

# Roll back source to a specific backup:
cp ~/projects/zmenu/.zmenu-backups/zmenu-20250325-143022.sh ~/projects/zmenu/zmenu.s

# Roll back the live binary directly (fastest recovery):
sudo cp ~/projects/zmenu/.zmenu-backups/zmenu-installed-20250325-143022.sh /usr/loca
sudo chmod +x /usr/local/bin/zmenu

# Confirm rollback:
grep 'ZMENU_VERSION' /usr/local/bin/zmenu
zmenu
```

Problem: Asking Claude to help — what to include

To save context and get a precise answer, give Claude:

1. **The exact function name(s)** you want changed (run `grep -n 'function_name' zmenu.sh`)
2. **The line range** of the relevant block (e.g. "lines 1342–1577")
3. **What you want it to do** — be specific
4. Tell Claude the **current version** so it knows what bump to apply

Claude will then read the live file, produce the full replacement block, and tell you exactly which lines to replace. You never need to paste the whole file.

QUICK COMMAND CARD

```bash
```

```bash
# Backup
ZMENU_SRC=~/projects/zmenu/zmenu.sh
cp "$ZMENU_SRC" ~/projects/zmenu/.zmenu-backups/zmenu-$(date +%Y%m%d-%H%M%S).sh

# Find a function
grep -n 'function_name' ~/projects/zmenu/zmenu.sh

# Edit
nano ~/projects/zmenu/zmenu.sh

# Syntax check
bash -n ~/projects/zmenu/zmenu.sh && echo "OK"

# Install
sudo cp ~/projects/zmenu/zmenu.sh /usr/local/bin/zmenu && sudo chmod +x /usr/local/b

# Verify + run
grep 'ZMENU_VERSION' /usr/local/bin/zmenu && zmenu

# Roll back
sudo cp ~/projects/zmenu/.zmenu-backups/zmenu-installed-TIMESTAMP.sh /usr/local/bin/
```

Keep this file at `~/projects/zmenu/EDIT-SOP.md` for quick reference.